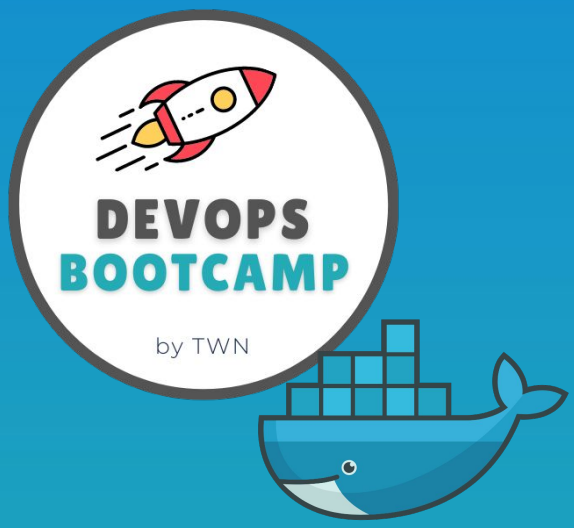


Module Checklist

Containers with Docker

By TechWorld with Nana



Video Overview

- ★ What is a Container?
- ★ Container vs Image
- ★ Docker (Container) vs Virtual Machine
- ★ Docker Architecture and its components
- ★ Main Docker Commands
- ★ Debug Commands
- ★ Demo Project: Overview
- ★ Demo Project: Developing with Docker
- ★ Demo Project: Docker Compose - Running multiple services
- ★ Demo Project: Dockerfile - Building our own Docker Image
- ★ Demo Project: Private Docker Repository - Pushing our Docker Image into a private Registry on Nexus
- ★ Demo Project: Deploying our containerized application
- ★ Docker Volumes - Persist data in Docker
- ★ Demo Project: Volumes - Configuring persistence for our application
- ★ Demo Project: Run Nexus as Docker container
- ★ Docker Best Practices

Check your progress... 1/6

What is a Container?

- ❑ Watched video

Container vs Image

- ❑ Watched video
- ❑ Installed Docker on your local machine
- ❑ **Demo executed** - run two different Versions of Postgres Docker Images

Useful Links:

- Docker Installation Guides for different OS: https://docs.docker.com/get-docker/
- Postgres Docker Images: https://hub.docker.com/_/postgres

Docker vs Virtual Machine

- ❑ Watched video

Docker components

- ❑ Watched video

Check your progress... 2/6

Main Docker Commands

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Pull Redis Docker Image (docker pull)
  - ❑ List existing Docker Images (docker images)
  - ❑ Run Container (docker run)
  - ❑ Run Container in a detached mode (docker run -d)
  - ❑ List running containers (docker ps)
  - ❑ Start container (docker start)
  - ❑ Stop container (docker stop)
  - ❑ List all containers - running and stopped ones (docker ps -a)
  - ❑ Bind port (docker run -p)

Useful Links:

- Redis Docker Images: https://hub.docker.com/_/redis

Debug Commands

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ See logs of container (docker logs)
  - ❑ Get interactive terminal of running container for troubleshooting (docker exec -it)

Demo Project: Overview

- ❑ Watched video

Check your progress... 3/6

Demo Project: Developing with Docker

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Git cloned example git project or created new one
  - ❑ Pulled mongodb image
  - ❑ Pulled mongo-express image
  - ❑ Created mongo-network
  - ❑ Started mongodb container with all necessary parameters
  - ❑ Started mongo-express container with all necessary parameters
  - ❑ Created new database via Mongo Express UI
  - ❑ Configured Nodejs application code to connect with database

Useful Links:

- MongoDB Docker Image: https://hub.docker.com/_/mongo
- Mongo Express Docker Image: https://hub.docker.com/r/techworldwithnana/mongo-express

Demo Project: Docker Compose - Running multiple services

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Installed Docker Compose (should already be installed with Docker Desktop)
  - ❑ Created a docker-compose file to start mongodb and mongo-express containers instead of using docker run
  - ❑ Created new database

Useful Links:

- Docker Compose Installation Guides for different OS: https://docs.docker.com/compose/install/

Check your progress... 4/6

Demo Project: Dockerfile - Building our own Docker Image

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Created Dockerfile for our Node application (the name of the file MUST be Dockerfile!)
  - ❑ Built Docker Image from our Dockerfile and tag it
  - ❑ Started newly created Docker Image

Demo Project: Private Docker Repository - Push/Pull to Nexus Repository

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Created a Docker Repository on Nexus
  - ❑ Created a User Role for Docker Repository on Nexus
  - ❑ Configured Repository Connector (port 8083)
  - ❑ Configured Firewall Rule to open port 8083 on Droplet
  - ❑ Configured Token Issuing on Nexus (Realm - activate Docker Bearer Token Realm)
  - ❑ Configured insecure registries for Nexus IP and Port in Docker Desktop (Docker Engine Tab)
  - ❑ Logged in to Nexus Docker Repo (docker login)
  - ❑ Pushed Docker Image to Nexus Repo
  - ❑ Fetched Docker Image from Nexus Repo

Check your progress... 5/6

Demo Project: Deploying our containerized application

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Added our example application to Docker compose file
  - ❑ Changed mongodb server url from localhost to mongodb service name in Node Code
  - ❑ Started docker containers with docker-compose

Docker Volumes - Persist data in Docker

- ❑ Watched video

Demo Project: Volumes - Configuring persistence for our application

- ❑ Watched video
- ❑ **Demo executed -** defined a Named Volume in Docker Compose File

Check your progress... 6/6

Demo: Project Run Nexus as Docker Container on DigitalOcean Droplet

- ❑ Watched video
- ❑ **Demo executed**
  - ❑ Created a new Droplet
  - ❑ Attached Droplet to existing Firewall
  - ❑ Installed Docker on Droplet
  - ❑ Created docker volume to persist Nexus data
  - ❑ Ran Nexus as Docker container with necessary parameters
  - ❑ Accessed Nexus in browser

Useful Links:

- Nexus Docker Image: https://hub.docker.com/r/sonatype/nexus3

More Resources...

Official Resources

- Best practices for writing Dockerfiles: https://docs.docker.com/build/building/best-practices/
- Docker development best practices: https://docs.docker.com/build/building/best-practices/
- Tips for Caching, reducing Image size, maintainability, reproducibility: https://www.docker.com/blog/intro-guide-to-dockerfile-best-practices/
- **Tip**: Enforce Dockerfile best practices automatically by using a static code analysis tool (e.g. https://github.com/hadolint/hadolint)

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